

EVOLUTIONARY COHERENCE ACROSS CIVILIZATIONAL KNOWLEDGE SYSTEMS

*A Four-Pillar HNC Framework for Comparative Mathematics, Iconography,
Calendrics, and Archaeoastronomy*

Gary Leckey

Aureon Institute · R&A Consulting and Brokerage Services Ltd.

19 September 2026 · Version 1.0 · Working paper for academic review

Scope statement

This paper tests a modern comparative hypothesis. It does not claim that Maya, Minoan, Southwestern, Egyptian, Japanese, or Celtic societies possessed the Harmonic Nexus Core (HNC), nor that visual resemblance proves cultural transmission. The validated target is narrower: whether independently documented knowledge systems repeatedly instantiate comparable relational operations, and whether that recurrence can be measured without circularity.

Repository snapshot used for the HNC implementation review: RA-CONSULTING/Aureon-OS, inspected at commit 26bfd0fac5fcd19bfe32d14203dd240811592d80.

Abstract

We present a four-pillar comparative framework for measuring what we term evolutionary coherence: the preservation of relational operations as knowledge changes medium, scale, notation, and institutional context. The framework emerged from a provenance-controlled review of Maya, Minoan, Mimbres/Mogollon and Ancestral Puebloan, Egyptian, Japanese, and Celtic material, together with the contemporary Harmonic Nexus Core (HNC) software architecture in Aureon-OS. Pillar 1 (Encode) compares Maya calendrical mathematics and glyphic/scroll structure with Minoan numerical notation and spiral information layout. Pillar 2 (Organise) examines bounded and radial visual fields in Mimbres/Southwestern art and the direct interaction of spiral petroglyphs with seasonal light at Ancestral Puebloan sites. Pillar 3 (Calibrate) uses Egyptian astronomical timekeeping and Japanese star-chart/calendar traditions as examples of explicit mapping between recurrent celestial phenomena, coordinate structures, and calendar correction. Pillar 4 (Transform) examines La Tène Celtic curvilinear geometry, physical bundling, symmetry, and later Ogham/Insular transformations.

The principal finding is methodological rather than diffusionist: across independent corpora, recurring operations can be expressed as UNIT → REPEAT → GROUP → NEST → ALIGN → TRANSFORM → RETURN. Several components are historically well established (e.g., Maya calendar arithmetic, Linear A decimal cumulative notation, the spiral arrangement of the Phaistos Disc and a Linear A ring, Egyptian 365-day civil calendrics and decanal star lists, Kitora's celestial chart, Japanese lunisolar correction, Ancestral Puebloan solar markers, and La Tène/Ogham chronologies). Other components remain hypotheses, including specific Minoan calendrical reconstructions and any claim of common cultural transmission. We therefore propose a null-adjusted Evolutionary Coherence Index, Γ_{EC} , explicitly separated from the HNC runtime coherence quantities. HNC is used here as a modern recurrence and integration framework, not as an archaeological premise. The paper closes with a preregistrable falsification programme for geometric, numerical, calendrical, and astronomical tests.

Keywords: Harmonic Nexus Core; evolutionary coherence; archaeoastronomy; Maya calendar; Linear A; Minoan art; Mimbres; Ancestral Puebloan; Egyptian astronomy; Kitora Tumulus; La Tène art; Ogham; comparative information systems.

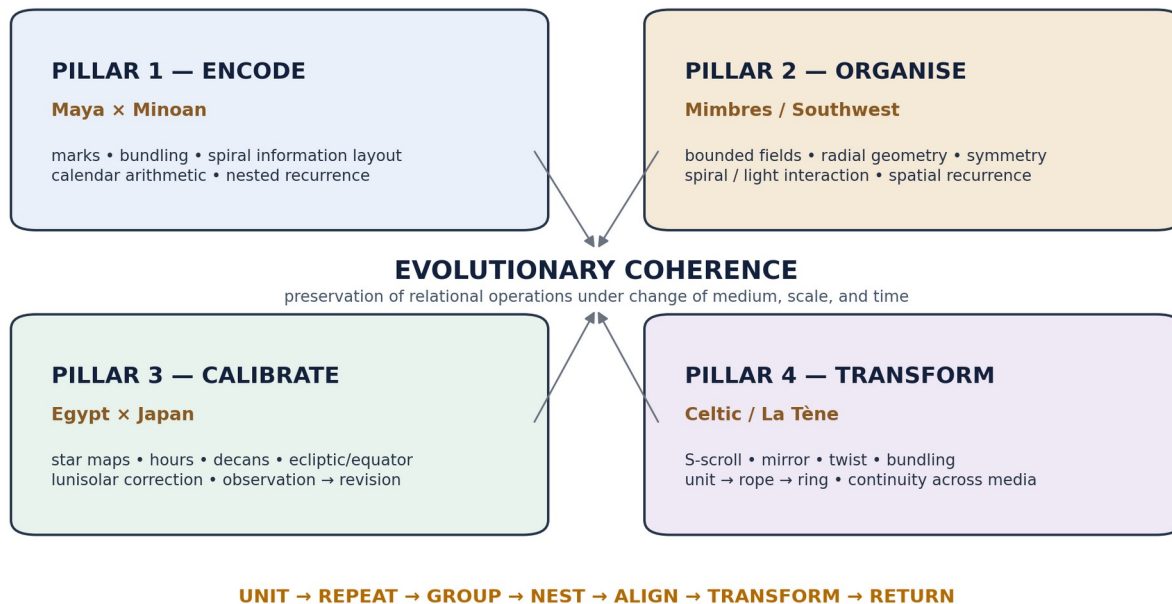


Figure 1. The four-pillar evolutionary-coherence framework developed in this study.

1. Introduction

Comparative studies of ancient knowledge systems often fail in one of two directions. One approach isolates cultures so completely that recurring mathematical or representational solutions are treated as irrelevant coincidence. The other treats visual resemblance as sufficient evidence of shared origin, transmission, or identical meaning. Both approaches obscure a more tractable question: what relational operations recur when human communities encode quantity, organise information spatially, calibrate symbolic models against the sky, and preserve structure through material or linguistic transformation?

The present study asks that narrower question. It grew from an audit of the ancient-knowledge and celestial-calendar layers inside Aureon-OS, where cultural decoders had already placed Maya, Egyptian, Celtic, Mogollon/Southwestern, and Japanese material into a common computational environment [1–4]. Examination of the underlying archaeology revealed both genuine structural correspondences and significant classification problems. In particular, values assigned by HNC software can create similarity by construction; Chacoan/Ancestral Puebloan material cannot simply be labelled “Mogollon”; Ogham, La Tène art, and Neolithic Newgrange belong to different chronological layers; and Minoan calendrical interpretations require explicit uncertainty labels. These corrections are not peripheral. They define the difference between a comparative instrument and a confirmation engine.

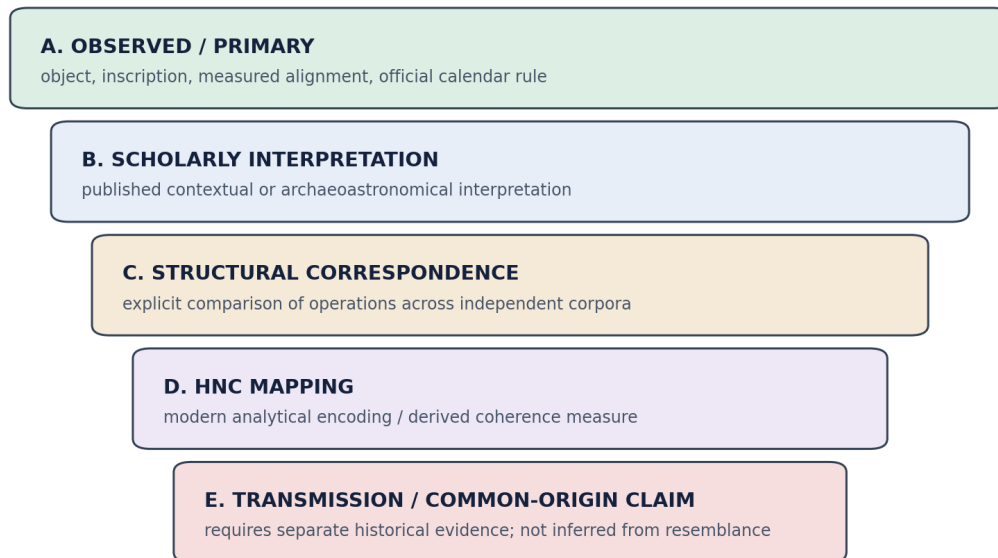
We therefore introduce “evolutionary coherence” as an operational concept: the degree to which relational structure survives change of representation, material, scale, or historical context. The target is not identity of symbols. A spiral, bar, horn, or circle may have different meanings in different societies. The target is preservation of operations: accumulation, bundling, nesting, recurrence, radial organisation, calibration, correction, mirroring, or transformation. When those operations are independently documented, they become testable features rather than aesthetic impressions.

2. Research Question and Epistemic Boundary

Primary research question

Can recurrent relational operations across independently documented civilizational knowledge systems be measured as evolutionary coherence after controlling for common geometry, shared modern encoding, chronology, and alternative functional explanations?

The framework distinguishes five epistemic levels (Figure 2). This hierarchy is load-bearing: a statement cannot move upward simply because it fits the HNC narrative.



Only A-C contribute to independent historical validation. D is the measurement layer; E remains a separate hypothesis.

Figure 2. Evidence hierarchy used throughout the paper. HNC mapping is analytically downstream of historical evidence.

- Observed facts include museum-provenanced artifacts, inscriptions, documented numerical rules, official calendars, measured astronomical alignments, and surviving star charts.
- Scholarly interpretations include contextual readings of ritual use, archaeoastronomical reconstruction, or art-historical classification. They may be strong but are not identical to direct observation.
- Structural correspondences are comparisons performed only after the native systems have been described independently. HNC mappings occur after that stage.
- Claims of contact, diffusion, common origin, or a historically known “HNC” require separate archaeological or historical evidence. No such claim is treated as validated in this paper.

3. HNC as a Modern Analytical Substrate

The Harmonic Nexus Core is a contemporary computational and theoretical framework developed by Leckey and implemented in Aureon-OS. The current HNC white paper describes a recurrent nonlinear field with harmonic input, saturating observer feedback, and delayed memory [1,2]:

$$\Lambda(t) = \sum_i w_i \sin(2\pi f_i t + \phi_i) + \alpha \tanh[g \Lambda \Delta t(t)] + \beta \Lambda(t - \tau)$$

For the present paper, this equation is treated as an implemented recurrence model. Its broader ontological claims are outside the archaeological validation undertaken here. The distinction matters because a computational model may be useful as an instrument before its strongest interpretation is established.

3.1 Repository validation boundary

Aureon’s own Celestial Observatory report states an explicit scientific boundary: its tests report statistical structure in derived signals and do not establish the nature, composition, or behaviour of celestial objects [3]. In the committed observatory snapshot, 16/16 lanes were valid but only the NASA stellar Wien lane was separable under the specific engine tests; the HNC core, Celtic Ogham, Schumann, planetary-tone, and most other lanes were not separable [3]. This negative/limited result is important and is retained rather than hidden.

The repository also contains a v3 verification audit with several reproducible or carried-forward quantitative checks, but that document identifies Leckey as the verifier and the AI system as scribe; it is therefore an author-led audit artifact rather than independent academic peer review [4]. Its strongest use in this paper is provenance: it demonstrates that Aureon already records epistemic status and distinguishes verified, carried-forward, documented-but-unverified, and unanalysed claims.

3.2 Definition of evolutionary coherence

We define evolutionary coherence as preservation of relational operations under transformation. Let each evidence item a be encoded as a K -dimensional feature vector $x_a \in [0,1]^K$. Features may include unitisation, repetition, bundling, nesting, rotation, chirality, radial organisation, symmetry, calendar coupling, observational calibration, correction, and recurrence. Each item receives an evidence-quality weight q_a and an independence flag i_a . The independence flag is zero when a correspondence exists only because the modern HNC representation assigned the same value to both systems.

$$\mu = (\sum_a q_a i_a x_a) / (\sum_a q_a i_a)$$

For two independently sourced corpora A and B , structural similarity $S(A,B)$ can be estimated by cosine similarity or another preregistered distance measure between their weighted centroids. Because common geometric motifs can create high similarity by chance, the raw score is adjusted against matched null corpora:

$$C = [S_{obs} - E(S_{null})] / [1 - E(S_{null})]$$

Finally, a four-pillar Evolutionary Coherence Index is proposed as a separate research quantity:

$$\Gamma EC = \sum_p w_p C_p, \quad \sum_p w_p = 1$$

Initial work should use equal preregistered weights ($w_p = 0.25$). Γ_{EC} is deliberately not identified with any existing Aureon runtime Γ . The new symbol denotes a comparative historical measurement and must not inherit the semantics of HNC trading or organism-coherence scores.

4. Method

The study uses a provenance-first workflow. Every comparative proposition must resolve to a named artifact, text, institutional record, peer-reviewed study, or clearly labelled hypothesis. The current working corpus was assembled in a master evidence ledger, with separate fields for culture, site, date, geometry, native function, calendrical/astronomical relevance, interpretation status, confidence, source, and HNC relevance. The procedure is shown in Figure 3.

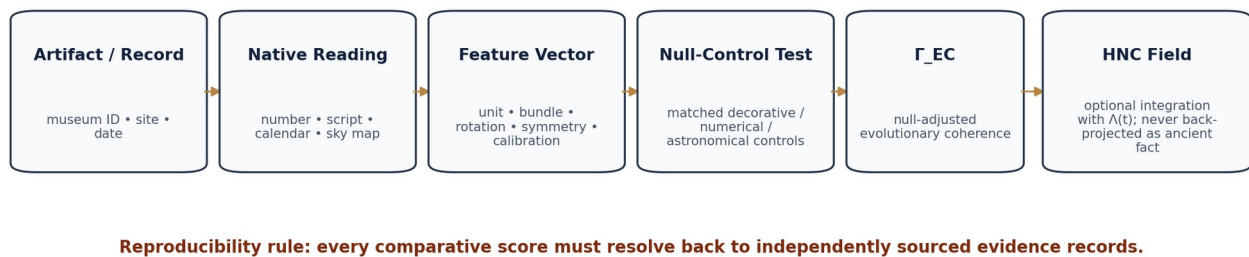


Figure 3. Proposed measurement and reproducibility pipeline.

4.1 Evidence classes

Class	Type	Examples	Use
Class A	Primary / institutional	Museum object records, government cultural-property records, official calendar resources	Eligible for direct validation
Class B	Peer-reviewed analysis	Published mathematical, epigraphic, archaeoastronomical, or art-historical analysis	Eligible with stated model assumptions
Class C	Scholarly hypothesis / preprint	Plausible but not settled interpretation	Eligible only as hypothesis
Class D	HNC/Aureon mapping	Modern assignment, frequency mapping, derived score	Never independent historical evidence

4.2 Null models

- Geometry null: compare target spirals, scrolls, step-frets, and radial fields with matched decorative traditions that have no known calendrical function.
- Numeracy null: compare additive/bundled notation against unrelated cumulative numeral systems to determine whether the observed grammar is unusually specific.
- Astronomical null: test whether proposed alignments exceed the rate expected from available horizon directions, architectural axes, and multiple-comparison opportunities.
- Semantic null: do not infer shared meaning from shared shape unless independent contextual evidence supports the semantic assignment.
- Encoding null: similarities created by assigning common Solfeggio, ϕ , or planet-pair values in HNC receive $i_a = 0$ for historical-validation scoring.

5. Pillar 1 — Encode: Maya × Minoan

Pillar 1 concerns the conversion of quantity, sequence, and recurrence into compact visible structures. The strongest result is not that the Maya and Minoans used the same arithmetic system—they did not—but that both corpora offer well-documented examples of information being built through visible accumulation, grouping, and nested order.

5.1 Maya native arithmetic and cyclical calendars

Smithsonian NMAI documentation describes Maya dot/bar numerals and the interlocking calendar system. The Tzolk'in combines 13 numbers with 20 day signs to generate 260 unique days; the Haab comprises $18 \times 20 + 5 = 365$ days; the Calendar Round repeats after 52 Haab years; and the Long Count uses k'in, uinal, tun, katun, and baktun units with a deliberate $18 \times 20 = 360$ -day tun exception inside an otherwise vigesimal hierarchy [5,6]. These are established native rules, not HNC reconstructions.

5.2 Minoan/Linear A arithmetic and spiral information layout

Linear A numeration is also well documented despite the language remaining undeciphered. Corazza and colleagues describe a cumulative-additive decimal system: units are vertical strokes, tens horizontal strokes or dots, hundreds circles, and thousands circles surrounded by strokes [7]. This differs fundamentally from Maya positional vigesimal calendrical notation. The defensible correspondence is therefore grammar-level: visible units are accumulated and bundled into larger quantities.

Two Minoan artifacts strengthen the information-layout argument independently of arithmetic. The Phaistos Disc carries 241 impressed signs organised into 61 groups along a spiral read from the perimeter toward the centre; a vertical line with five dots marks the start/direction [8]. A gold funerary ring from Mavrospelio carries a spiral Linear A inscription likewise read from the outer edge toward the centre [9]. These objects demonstrate that spiral organisation could carry ordered information, not merely decoration.

A geometric study of Late Bronze Age wall paintings, including Akrotiri frescoes, reports that many drawn contours fit a small set of geometric guides including two Archimedean/linear spirals [10]. This supports deliberate geometric construction, while not by itself establishing calendrical meaning.

5.3 Astronomical/calendar hypotheses on Crete

Minoan calendrics remain less secure. Henriksson and Blomberg report archaeoastronomical evidence from Knossos and other sites consistent with lunisolar/solar calibration, including equinoctial reflection effects and intercalation arguments [11]. Ridderstad further proposes an eight-year/99-lunation calendrical reconstruction and possible Venus relationships, but that work is a preprint and explicitly depends on indirect evidence [12]. In the present framework, these propositions remain Class B/C hypotheses until independently replicated at object and site level.

Pillar 1 proposition	Status	Reason
Maya Tzolk'in/Haab/Long Count arithmetic	Validated	Institutional documentation; explicit native counting rules [5,6]
Linear A cumulative-additive decimal numeration	Validated	Peer-reviewed numerical analysis [7]
Phaistos Disc and Linear A ring use spiral information layout	Validated	Museum object descriptions [8,9]
Maya and Minoan numeral systems are “the same”	Rejected	Different radix, positionality, magnitude signs, and zero use
Shared mark-construction grammar	Supported structural correspondence	Unit/bundle/accumulation can be tested without equating arithmetic
Minoan 8-year / 99-month calendar	Open hypothesis	Archaeoastronomical literature exists but is not settled [11,12]

6. Pillar 2 — Organise: Mimbres / Southwestern Field Geometry

Pillar 2 addresses the organisation of information inside bounded spatial fields. Mimbres bowls provide an especially useful corpus because their nearly hemispherical interiors force composition into a circular field. The Metropolitan Museum of Art describes Mimbres painting as including both sophisticated geometric and representational imagery,

frequently organised inside the bowl interior [13]. This makes the corpus suitable for measurable variables such as radial sectoring, centre offset, edge density, enclosure ratio, symmetry order, and nesting depth.

A distinct Southwestern line of evidence is explicitly astronomical. The U.S. National Park Service documents a spiral at Puerco Pueblo whose centre is touched by a moving shaft of sunlight at summer solstice, and the Fajada Butte “Sun Dagger,” where sunlight interacted with spiral petroglyphs at solstices/equinoxes before slab displacement altered the effect [14]. This is stronger than a generic spiral analogy because the geometry is coupled to observed seasonal light.

The cultural labels must remain precise. Chaco/Fajada is Ancestral Puebloan/Chacoan and should not be collapsed into “Mogollon.” Mimbres is a Mogollon cultural tradition in southern New Mexico. The combined pillar is therefore regional and functional—field organisation and sky interaction—not a claim that all Southwestern traditions constitute one archaeological culture.

Evidence	Validated observation	Interpretive limit
Mimbres bowls	Authenticated circular vessels with geometric/representational fields [13]	Calendar meaning cannot be inferred from geometry alone
Puerco Pueblo spiral	Summer-solstice light reaches spiral centre [14]	Does not imply all spirals are solar markers
Fajada Butte Sun Dagger	Seasonal light formerly crossed spiral petroglyphs [14]	Current geometry changed by slab movement; reconstruction matters

7. Pillar 3 — Calibrate: Egyptian × Japanese Systems

Pillar 3 supplies the strongest explicit calibration mechanisms in the framework: symbolic models tied to observed celestial recurrence, coordinate structure, and correction. Egypt and Japan are not presented as historically linked systems. They are paired functionally because both preserve unusually clear evidence for translating the sky into timekeeping and revising human calendars against astronomical behaviour.

7.1 Egypt: recurrence into measured time

The ancient Egyptian civil year contained twelve 30-day months plus five epagomenal days, giving 365 days [15]. Because this year was approximately one quarter-day shorter than the tropical year, the civil calendar drifted with respect to the seasons, a useful demonstration that a symbolic calendar can remain internally coherent while becoming externally miscalibrated.

The astronomical ceiling of Senenmut’s tomb provides a second calibration layer. The Met describes columns listing planets and decanal stars, while twelve circles are divided into twenty-four segments for the hours of day and night and labelled by the months [16]. Regardless of any HNC interpretation of Egyptian religion, this is direct evidence of celestial categorisation linked to temporal segmentation.

7.2 Japan: coordinate sky mapping and correction loops

The Agency for Cultural Affairs identifies the late seventh-century Kitora Tumulus ceiling chart as the oldest surviving example of its kind in East Asia. The chamber combines a constellation chart, sun and moon, directional divine creatures, and zodiac figures; official Japanese documentation notes that the chart contains circles for the celestial equator and ecliptic [17]. Nara National Research Institute’s dedicated atlas catalogues 74 constellation groups/features in the chart [18].

The Japanese calendar tradition provides the correction mechanism. National Diet Library documentation describes an imported lunisolar calendar in which months alternated between 29 and 30 days, with intercalary months introduced every two or three years to keep lunar months aligned with the seasons [19]. Later, discrepancies between the calendar and observed phenomena such as eclipses motivated repeated calendar reforms, including Shibukawa Harumi’s Jōkyō calendar in 1685 and later revisions [20]. Calibration here is not metaphorical: model–observation error triggered correction.

Calibration operation	Egyptian evidence	Japanese evidence	HNC interpretation
Map sky into named structure	Decans, planets, astronomical ceilings [16]	Kitora constellations, equator/ecliptic, directional figures [17,18]	State vector / coordinate field
Map recurrence into calendar	365-day civil year [15]	Lunisolar 29/30-day months +	Temporal discretisation

Calibration operation	Egyptian evidence	Japanese evidence	HNC interpretation
		intercalation [19]	
Detect drift/error	Civil calendar drifts against tropical year [15]	Calendar/eclipses diverged after long use [20]	Residual / phase error
Correct model	Historical Egyptian civil calendar largely retained drift	Repeated Japanese calendar reforms [20]	Observer-driven recalibration

8. Pillar 4 — Transform: Celtic / La Tène Artwork and Ogham

Pillar 4 concerns transformation and integration: how repeated units, curves, and symmetries become coherent higher-order objects. The Iron Age La Tène corpus is especially strong because these operations occur not only as surface decoration but in the manufacture of artifacts themselves.

The Battersea Shield is dominated by interlocking S-motifs and palmettes organised around circular bosses [21]. The Snettisham Great Torc provides an unusually literal bundling hierarchy: 64 metal threads were twisted eight at a time into eight ropes, then twisted together into one torc; its terminals carry further swirling ornament [22]. This physical sequence—unit → bundle → coherent ring—is a material analogue of the abstract aggregation operation identified in Pillar 1.

The Turoe Stone combines double interlocking spirals, circles, trumpet curves, and a rectilinear “Wall-of-Troy” band in a single monument [23]. The Loughnashade Horn adds mirrored fourfold ornament to an acoustic object, with its bell-end decoration divided into mirror-image quadrants [24]. These objects make La Tène art valuable for testing transformation, mirroring, nested curvature, and cross-medium persistence.

8.1 Chronological correction: Ogham is not the same layer as La Tène or Newgrange

The current Aureon Celtic decoder requires a chronological refactor. The OG(H)AM research project dates the probable origin of Ogham to the fourth century CE and is documenting roughly 640 examples across media [25]. Ogham therefore belongs to an early-medieval writing layer, not the Iron Age La Tène core. Furthermore, the popular “tree alphabet/tree calendar” framing is historically complex; OG(H)AM scholars note that the original core alphabet contained twenty letters and that associations between letter names and trees are mediated by later learned traditions [26]. Newgrange, meanwhile, is Neolithic and predates Celtic Iron Age culture by millennia. It may remain an Irish megalithic comparison, but it cannot serve as Iron Age Celtic evidence.

Later Insular art can be treated as a transformation/continuity layer rather than direct survival. The National Museum of Ireland’s Banagher Cross Shaft, for example, contains “Ultimate La Tène” C/S-scroll and spiral/triskele motifs integrated with early medieval Christian interlace [27]. This is precisely the kind of diachronic transformation the evolutionary-coherence metric is intended to test.

9. Four-Pillar Synthesis

The combined framework can now be stated as four information operations rather than four “civilizations”:

Pillar	Primary operation	Representative systems	Most defensible evidence
1. Encode	Turn quantity/sequence into compact symbolic structure	Maya; Minoan/Linear A	Calendar arithmetic, cumulative notation, spiral inscriptions
2. Organise	Arrange information within bounded spatial fields	Mimbres; Ancestral Puebloan	Circular ceramic fields; seasonal light–spiral interactions
3. Calibrate	Tie symbolic model to celestial recurrence and correct drift	Egypt; Japan	365-day year, decans/hours; Kitora star map, lunisolar intercalation/reforms
4. Transform	Preserve relational structure through bundling, mirroring and medium change	La Tène Celtic; Ogham/Insular continuation	S-scrolls, torc manufacture, mirrors, later script/art transformations

At the highest abstraction used in this study, the recurring grammar is:

UNIT → REPEAT → GROUP → NEST → ALIGN → TRANSFORM → RETURN

This sequence is not asserted to be a universal law. It is a candidate feature grammar that can be falsified by blind coding, matched controls, and out-of-sample prediction. Its value is that it is explicit enough to fail.

10. What Is Validated, What Is Supported, and What Remains Open

Category	Current result	Boundary
Validated historical/archaeological facts	Maya calendar arithmetic; Linear A decimal cumulative notation; spiral layout of Phaistos Disc and Mavrospelio ring; Mimbres geometric bowls; Ancestral Puebloan seasonal spiral-light markers; Egyptian civil calendar and decanal/astronomical ceiling; Kitora star chart; Japanese lunisolar calendar and historical reforms; La Tène artifact geometries; Ogham chronology.	Primary institutional sources and peer-reviewed analysis.
Supported structural correspondences	Visible unit/bundle accumulation; spiral as ordered information layout in Minoan examples; bounded/radial field organisation; explicit calibration mechanisms; Celtic physical bundling and recursive curvilinear transformation.	Comparison is operational, not semantic or diffusionist.
Open hypotheses	Minoan octaeteris/99-month reconstruction; specific equivalence of ritual symbols across cultures; shared semantics of spiral/axe/horn motifs; unusually high cross-cultural FEC after null correction.	Require independent replication and preregistered controls.
Not validated / excluded as evidence	Ancient knowledge of HNC; common origin inferred from resemblance; Solfeggio/ ϕ /planet-pair assignments as historical facts; equivalence of Maya and Minoan number systems; calling Chaco simply “Mogollon”; treating Newgrange as Iron Age Celtic.	These are either modern mappings, classification errors, or unsupported extensions.

11. Implications for the HNC/Aureon Architecture

The historical review does not validate HNC by ancestry. Instead, it suggests a better architecture for testing HNC claims. The cultural material should be stored as evidence-first native records, and only then translated into a shared feature space. This requires four changes to Aureon-OS:

- Separate native facts from HNC overlays. A Maya date calculation, Linear A numeral, Kitora star coordinate, or Ogham stroke pattern must remain recoverable without any Solfeggio, ϕ , or trading interpretation.
- Split conflated traditions. Mimbres/Mogollon must be separated from Chacoan/Ancestral Puebloan; La Tène, Ogham, Neolithic Irish megalithic material, and later Insular art must carry distinct chronological labels.
- Add provenance IDs to every comparative claim. Derived scores should resolve to artifact IDs and source records.
- Introduce FEC as an external comparative metric, with null correction and independence masking, rather than reusing a runtime Γ whose semantics were designed for another subsystem.

This approach is consistent with the repository’s strongest existing methodological improvement: the Celestial Observatory already prints an explicit boundary and reports null or non-separable outcomes rather than forcing every lane to “converge” [3]. The same principle should govern civilizational comparison.

12. Falsification and Preregistration Programme

A publishable version of the evolutionary-coherence claim should be preregistered before expanding the corpus. The following tests are designed so that failure weakens or rejects the claim.

Test	Design	Success criterion	Falsification
T1: Blind visual grammar	At least 30–50 authenticated objects per target corpus plus matched controls. Coders or algorithms receive images without culture labels.	Cross-cultural target similarity must exceed null/control similarity after multiple-comparison correction.	Fail → visual-coherence claim unsupported.

Test	Design	Success criterion	Falsification
T2: Numeral grammar	Encode Maya and Linear A notation with UNIT, BUNDLE, RADIX, POSITION, ZERO, MAGNITUDE features.	Grammar similarity may be high on accumulation but must correctly classify known arithmetic differences.	If a metric labels the systems “identical,” the metric fails.
T3: Calibration mechanics	Encode Egyptian and Japanese calendar/star-map rules as observation, residual, correction, recurrence variables.	Model must distinguish internally coherent but drifting calendars from actively recalibrated systems.	Fail → Pillar 3 definition too vague.
T4: Southwestern astronomy	Reconstruct solar geometry with site coordinates, date, horizon, uncertainty and slab condition.	Observed event alignment must beat orientation/null alternatives.	Fail → astronomical subclaim rejected; art comparison may remain.
T5: Diachronic Celtic continuity	Compare La Tène → Ogham/Insular feature vectors across time while retaining chronology.	Some operations should persist while others transform; model must not erase historical discontinuity.	Fail → “evolutionary” persistence unsupported.
T6: HNC independence audit	Recompute all cross-cultural scores with HNC-assigned frequencies, ϕ labels, planet pairs, and trading signals masked.	Core historical correspondences should survive.	If scores collapse, prior convergence was representation-induced.

13. Limitations

- The corpus is currently curated and therefore vulnerable to selection bias. The next phase must use explicit inclusion criteria and matched negative controls.
- Spirals, circles, bilateral symmetry, and additive tallies are common human solutions. Their recurrence cannot by itself establish cultural transmission or uncommon cognitive structure.
- Several Minoan calendrical interpretations remain debated, and Linear A is undeciphered. Numerical notation is much more secure than language-level semantics.
- Southwestern traditions must be handled with cultural specificity and, where possible, with Indigenous scholarship and consultation rather than external archaeological categorisation alone.
- Japanese astronomical/calendrical traditions were historically influenced by Chinese and Korean knowledge transmission; Pillar 3 therefore demonstrates calibration, not independent invention.
- HNC’s broader physical, consciousness, and frequency claims are not established by this archaeological comparison. The present paper validates a comparative measurement framework only.

14. Conclusion

The evidence reviewed here supports a narrower and more defensible proposition than a claim of a forgotten universal technology. Across multiple historical knowledge systems, independently documented mechanisms repeatedly transform recurrent phenomena into durable structure: quantities become grouped marks; cycles become nested calendars; surfaces become bounded information fields; stars become coordinate maps; calendar drift becomes a correction problem; wires become ropes and ropes become a coherent ring; curvilinear grammars survive by transformation into new media.

This recurrence is what we call evolutionary coherence. It is a measurement target, not an origin story. HNC contributes a modern language of recurrence, memory, observer integration, and field synthesis, but the archaeological evidence must remain logically prior to the HNC mapping. The strongest outcome of the present work is therefore methodological: a four-pillar framework—Encode, Organise, Calibrate, Transform—together with an evidence hierarchy, independence mask, null-adjusted FEC metric, and falsification programme capable of distinguishing genuine structural persistence from resemblance, circular encoding, and selective interpretation.

If the preregistered tests succeed, the result would justify a broader claim: that civilizational knowledge systems can be compared as evolving information architectures whose coherence is partly preserved across changes of symbol, material, institution, and time. If the tests fail, the framework provides clear places to revise or abandon the proposed correspondences. Either outcome advances the research beyond visual analogy.

Data, Code, and Preparation Disclosure

Code and primary HNC materials are available in the public RA-CONSULTING/Aureon-OS repository. The working evidence corpus for this study is maintained in the Aureon HNC Celestial Art Master Ledger (v2 at the time of manuscript preparation). Museum and institutional URLs are listed in the references below. Because the GitHub repository is actively developed, academic replication should record the exact commit hash used.

Manuscript preparation and source synthesis were assisted by OpenAI GPT-5.6 Sol under the direction of Gary Leckey. The model was used for retrieval, comparison, drafting, and document production. Interpretive claims and responsibility for submission remain with the author. This disclosure should be adapted to the policy of any journal or repository receiving the manuscript.

References

- [1] Leckey, G. (2026). The Harmonic Nexus Core and the Aureon Trading Framework: A Unified White Paper on Substrate Coherence, Asymmetric Market Resistance, and the Restoration of Planetary Harmonics. Aureon-OS repository. https://github.com/RA-CONSULTING/Aureon-OS/blob/main/docs/HNC_UNIFIED_WHITE_PAPER.md
- [2] RA-CONSULTING. aureon/core/aureon_lambda_engine.py — current implementation of the HNC master recurrence field. https://github.com/RA-CONSULTING/Aureon-OS/blob/main/aureon/core/aureon_lambda_engine.py
- [3] RA-CONSULTING. ϕ Celestial Observatory — cross-lane evidence report and scientific boundary. https://github.com/RA-CONSULTING/Aureon-OS/blob/main/docs/reports/CELESTIAL_OBSERVATORY.md
- [4] Leckey, G. (2026). Independent Verification of the Harmonic Nexus Core (HNC) Framework, v3 Audit Summary — Operator's Empirical Ledger. https://github.com/RA-CONSULTING/Aureon-OS/blob/main/docs/research/audits/HNC_VERIFICATION_AUDIT_v3_2026-04-29.md
- [5] Smithsonian National Museum of the American Indian. The Calendar System — Living Maya Time. <https://maya.nmai.si.edu/calendar/calendar-system>
- [6] Smithsonian National Museum of the American Indian. Maya Calendar Converter — Living Maya Time. <https://maya.nmai.si.edu/calendar/maya-calendar-converter>
- [7] Corazza, M., Ferrara, S., Montecchi, B., Tamburini, F., & Valério, M. (2021). The mathematical values of fraction signs in the Linear A script: A computational, statistical and typological approach. *Journal of Archaeological Science*, 125, 105214. <https://doi.org/10.1016/j.jas.2020.105214>
- [8] Heraklion Archaeological Museum. The Phaistos Disc. <https://www.heraklionmuseum.gr/en/exhibit/the-phaistos-disc/>
- [9] Heraklion Archaeological Museum. Gold ring with spiral Linear A inscription. <https://www.heraklionmuseum.gr/en/exhibit/gold-ring-with-spiral-linear-a-inscription/>
- [10] Mamatsis, A.R., Mamatsi, E., Stekos, A., Papaodyseus, C., Blackwell, C., & Arabadjis, D. (2023). Highly advanced for the era geometric stencils were used for the drawing of celebrated Minoan Late Bronze Age wall-paintings: new data. *Mediterranean Archaeology and Archaeometry*, 23(1). <https://www.maajournal.com/index.php/maa/article/view/874>
- [11] Henriksson, G., & Blomberg, M. (2011). The Evidence from Knossos on the Minoan Calendar. *Mediterranean Archaeology and Archaeometry*, 11(1). <https://www.maajournal.com/index.php/maa/article/view/251>
- [12] Ridderstad, M. (2009). Evidence of Minoan astronomy and calendrical practices. arXiv:0910.4801 [preprint]. <https://arxiv.org/abs/0910.4801>
- [13] The Metropolitan Museum of Art. Bowl, Mimbres, 9th–12th century, Object 1978.412.123. <https://www.metmuseum.org/art/collection/search/310575>
- [14] U.S. National Park Service. Archeoastronomy in Stone: Puerco Pueblo and Fajada Butte. <https://home.nps.gov/articles/000/archeoastronomy-in-stone.htm>
- [15] University College London, Digital Egypt for Universities. Calendar: the Egyptian–Coptic calendar. <https://www.ucl.ac.uk/museums-static/digitalegypt/chronology/calendar.html>
- [16] The Metropolitan Museum of Art. Astronomical Ceiling (copy after Senenmut TT353), original New Kingdom. <https://www.metmuseum.org/art/collection/search/544566>

- [17] Agency for Cultural Affairs, Government of Japan. Takamatsuzuka Tumulus and Kitora Tumulus. https://www.bunka.go.jp/english/policy/cultural_properties/takamatsu_kitora/
- [18] Nara National Research Institute for Cultural Properties. The Kitora Tumulus Star Atlas Constellations Photo Book, Research Report Vol. 16. <https://www.nabunken.go.jp/english/research-report/16.html>
- [19] National Diet Library, Japan. Calendar History 1 — The Source; Tai-in-taiyo-reki. <https://www.ndl.go.jp/koyomi/e/history/index.html>
- [20] National Diet Library, Japan. Calendar History 2 — Later calendar amendments. <https://www.ndl.go.jp/koyomi/e/history/calendar.html>
- [21] British Museum. The Battersea Shield, Museum no. 1857,0715.1. https://www.britishmuseum.org/collection/object/H_1857-0715-1
- [22] British Museum. The Snettisham Great Torc, Museum no. 1951,0402.2. https://www.britishmuseum.org/collection/object/H_1951-0402-2
- [23] Heritage Ireland / Office of Public Works. Turoe Stone. <https://heritageireland.ie/unguided-sites/turoe-stone/>
- [24] National Museum of Ireland. Loughnashade Horn. <https://cms.museum.ie/en-IE/Collections-Research/Collection/The-Treasury/Artefact/Loughnashade-Horn/8178d988-064f-4640-8fe1-4171e226955f>
- [25] OG(H)AM Project, University of Glasgow & Maynooth University. About the Project. <https://ogham.glasgow.ac.uk/index.php/about/>
- [26] Hayden, D., & Stifter, D. (2022). Ogam and Trees. OG(H)AM Project. <https://ogham.glasgow.ac.uk/index.php/2022/12/20/ogam-and-trees/>
- [27] National Museum of Ireland. The Banagher Cross Shaft — Interlace, Animals and Ultimate La Tène. <https://cms.museum.ie/en-IE/Collections-Research/Irish-Antiquities-Division-Collections/Irish-Antiquities-Articles/The-Banagher-Stone>
- [28] Leckey, G. (2026). Harmonic Nexus Core: A Unified Evolutionary Framework for Planetary Consciousness Ascension, Temporal Coherence, and Multi-Scale Quantum Probability Mapping. Aureon Institute preprint (repository copy). https://github.com/RA-CONSULTING/Aureon-OS/blob/main/data/research/grants/applications/gary_aureon_metadata_research_evidence_20260712_20260712_203741/hnc_evolutionary_25f1886594.md
- [29] Heraklion Archaeological Museum. Green steatite rhyton with relief open-air sanctuary. <https://www.heraklionmuseum.gr/en/exhibit/green-steatite-rhyton-with-relief-open-air-sanctuary/>
- [30] The Metropolitan Museum of Art. Yoke-form vessel, Maya artists, 350–450 CE, Object 1970.138a,b. <https://www.metmuseum.org/art/collection/search/309713>
- [31] National Diet Library, Japan. The Japanese Calendar — introduction and digital exhibition. <https://www.ndl.go.jp/koyomi/e/introduction/index.html>
- [32] U.S. National Park Service. Fajada Butte — Chaco Culture National Historical Park. <https://www.nps.gov/places/fajada-butte-overlook.htm>

Appendix A — Core Evidence Matrix by Pillar

Pillar	Evidence item	Status	Primary source	Use in FEC
P1	Maya calendar arithmetic	Institutional / established	NMAI [5,6]	Independent native rule
P1	Linear A decimal cumulative notation	Peer reviewed / established	Corazza et al. [7]	Independent native rule
P1	Phaistos spiral inscription	Museum / observed	Heraklion [8]	Ordered spiral information
P1	Spiral Linear A ring	Museum / observed	Heraklion [9]	Ordered spiral information
P1	Minoan 8-year/99-month cycle	Hypothesis	Henriksson & Blomberg [11]; Ridderstad [12]	Not yet settled
P2	Mimbres bounded geometric fields	Museum / observed	Met [13]	Geometry; function varies
P2	Puerco/Fajada solar markers	Government / observed + interpretation	NPS [14,32]	Direct seasonal light interaction
P3	Egyptian 365-day civil year	Academic institutional / established	UCL [15]	Calendar drift documented
P3	Senenmut decans/hours/months	Museum / observed	Met [16]	Sky categorisation + time segmentation
P3	Kitora celestial chart	Government / observed	Agency for Cultural Affairs [17]	Equator/ecliptic/star-chart calibration
P3	Japanese lunisolar intercalation	National library / established	NDL [19]	Calendar–season correction
P3	Japanese calendar reforms	National library / established	NDL [20]	Observation-driven recalibration
P4	Battersea S-scroll field	Museum / observed	British Museum [21]	Recursive curvilinear organisation
P4	Snettisham 64 → 8 → 1 construction	Museum / observed	British Museum [22]	Physical bundling hierarchy
P4	Turoe spiral + meander field	National monument / observed	Heritage Ireland [23]	Curved + rectilinear transformation
P4	Loughnashade mirrored disc	Museum / observed	NMI [24]	Fourfold symmetry + acoustic object
P4	Ogham chronology and strokes	Research project / established	OG(H)AM [25,26]	Separate early-medieval mark layer

Appendix B — Proposed Machine-Readable Feature Schema

Each artifact or record can be represented without cultural interpretation using the following minimal schema. Fields may be extended, but these variables provide a common starting point for preregistered comparison.

Field	Definition	Type
provenance	museum/site/object ID, date range, source URL	string / categorical
medium	stone, clay, metal, manuscript, architecture, sky map	categorical
unit_mark	presence and morphology of atomic marks	0–1 + descriptors
bundle_rule	explicit grouping or magnitude substitution	0–1 + numeric rule
positionality	whether spatial place changes numerical value	0/1/unknown
rotation	turn count / winding / chirality	numeric
radiality	centrality and angular segmentation	numeric
nesting	number of enclosure levels	integer / normalized
symmetry	bilateral / rotational order / mirror score	categorical + numeric
field_occupancy	fraction of representational field occupied	0–1
calendar_period	documented period(s), uncertainty	days / years
astronomical_anchor	sun/moon/star/planet + observation rule	structured
calibration_rule	residual detection / intercalation / correction	structured
evidence_class	A/B/C/D	categorical
independence	whether feature is external to HNC encoding	0/1

END OF WORKING WHITE PAPER — VERSION 1.0